

CM5465 CLOUD COMPUTING

UNIT1

2-Marks Questions (Remembering Level)

1. Define cloud computing.
2. List any two essential characteristics of cloud computing.
3. What is a public cloud?
4. Define IaaS in cloud computing.
5. What is a hybrid cloud?
6. Mention any two benefits of cloud computing.
7. What is SaaS? Give one example.
8. Define cloud deployment model.
9. What is meant by cloud infrastructure?
10. List any two components of cloud computing architecture.

4-Marks Questions (Remembering & Understanding Level)

1. Explain any four essential characteristics of cloud computing.
2. Differentiate between public and private cloud.
3. Explain IaaS with a suitable example.
4. Describe any two cloud service models with examples.
5. Write a short note on cloud computing benefits.
6. Explain cloud computing infrastructure with a diagram.
7. Describe the architecture of cloud computing.
8. What is a cloud-based Integrated Development Environment (IDE)? Explain with an example.
9. Compare community cloud and hybrid cloud.
10. Explain the economic benefits of cloud computing.

6-Marks Questions (Application Level)

1. How would you apply cloud computing to a real-world scenario like online video streaming services?
2. Consider an educational institution planning to adopt cloud computing. Which cloud deployment model would be suitable and why?
3. Suppose a startup is developing a new mobile application. Explain how PaaS would help in the development process.
4. Analyze the impact of cloud computing on software development using cloud-based IDEs.

5. If a company needs scalable and cost-effective infrastructure, which cloud service model should they choose? Justify your answer.
6. Design a cloud computing architecture for an e-commerce platform. Explain the essential components.
7. A healthcare organization needs to store patient data securely. Which cloud deployment model should they adopt and why?
8. Explain how cloud computing can help a small business reduce IT costs.
9. Identify the challenges faced while implementing cloud infrastructure in an enterprise. Propose solutions for overcoming these challenges.
10. Compare and contrast traditional software development environments with cloud-based IDEs.

UNIT2

Here are the questions categorized based on cognitive levels and marks distribution:

2-Marks Questions (Remembering Level)

1. Define virtualization.
2. What is a Virtualization Reference Model?
3. List any two characteristics of a virtualized environment.
4. Name any two types of virtualization.
5. What is VMware?
6. Mention any two advantages of using virtual machines (VMs).
7. Define VM migration.
8. List any two virtualization technologies.
9. What is the role of a hypervisor in virtualization?
10. Mention any two disadvantages of virtualization.

4-Marks Questions (Remembering & Understanding Level)

1. Explain the characteristics of a virtualized environment.
2. Describe the Virtualization Reference Model with a diagram.
3. Differentiate between full virtualization and para-virtualization.
4. Explain any two types of virtualization with examples.
5. Compare VMware and Microsoft Hyper-V.
6. What is VM consolidation? Explain with an example.
7. Discuss VM migration and its benefits.
8. Explain how KVM and Xen differ from each other.
9. Describe any four advantages of virtualization.
10. What are the disadvantages of virtualization? Explain with examples.

6-Marks Questions (Application Level)

1. Suppose an organization wants to optimize its IT infrastructure using virtualization. Suggest a suitable virtualization technology and justify your choice.
2. A company is planning to migrate its existing servers to a virtualized environment. Explain the steps and benefits of VM migration.
3. How would you apply virtualization in a cloud computing environment to improve scalability and resource utilization?
4. Consider a scenario where multiple low-utilization physical servers exist. How can VM consolidation help in improving efficiency?
5. Analyze the impact of virtualization on data center operations with respect to cost savings and energy efficiency.
6. Suppose an IT company needs to set up a testing environment for software development. Which type of virtualization would you recommend and why?
7. Compare and contrast VMware and KVM in terms of performance, scalability, and ease of management.
8. Discuss how virtualization contributes to disaster recovery and business continuity.
9. Identify the security challenges associated with virtualization and propose solutions to mitigate them.
10. Evaluate the limitations of virtualization in high-performance computing applications.

UNIT3

2-Marks Questions (Remembering Level)

1. Define storage system architecture.
2. What is a Virtualized Data Centre (VDC)?
3. List any two benefits of a Virtualized Data Centre.
4. What is server virtualization?
5. Define storage virtualization.
6. What is the difference between block-level and file-level storage virtualization?
7. What is Virtual Provisioning in storage?
8. Mention two benefits of Virtual Storage Area Network (VSAN).
9. What is Google File System (GFS)?
10. Define Hadoop Distributed File System (HDFS).

4-Marks Questions (Remembering & Understanding Level)

1. Explain the architecture of a storage system.
2. Describe the components of a Virtualized Data Centre (VDC) architecture.
3. Differentiate between server virtualization and desktop virtualization.
4. Explain block-level storage virtualization with an example.

5. Compare Virtual Provisioning and Automated Storage Tiering.
6. What is a Virtual Storage Area Network (VSAN)? Explain its advantages.
7. Describe the working of Google File System (GFS).
8. Explain how Hadoop Distributed File System (HDFS) manages large-scale data storage.
9. Discuss networking virtualization and its benefits in a Virtualized Data Centre.
10. Explain how storage and application virtualization enhance cloud computing efficiency.

6-Marks Questions (Application Level)

1. Suppose an organization wants to improve its data center efficiency. How can virtualization techniques be applied to achieve this?
2. A company is struggling with high storage costs. How can virtual provisioning and automated storage tiering help optimize storage usage?
3. Design a Virtualized Data Centre (VDC) for a mid-sized enterprise, including storage, networking, and application virtualization.
4. Explain how a Virtual Storage Area Network (VSAN) can improve storage management in a cloud-based enterprise.
5. Analyze the role of Google File System (GFS) in handling large-scale distributed storage systems.
6. If a research institute needs to store and process large amounts of unstructured data, how would HDFS be beneficial?
7. Compare traditional file storage methods with block-level and file-level storage virtualization. Which would you recommend for cloud computing?
8. Discuss the security challenges in a Virtualized Data Centre and propose solutions to mitigate them.
9. A cloud service provider wants to implement efficient storage solutions. Suggest an approach using storage virtualization techniques.
10. Consider a scenario where a company wants to migrate from traditional storage to a cloud-based storage system. What factors should they consider, and which cloud file system (GFS or HDFS) would be most suitable?

Unit4

Remembering Level Questions (2 Marks each)

1. Define a cloud service provider.
2. What is a cloud service user?
3. List any two types of Service Level Agreements (SLA).
4. What is a Service Catalog?
5. Mention any two functions of a cloud portal.
6. Name the four phases of the Cloud Service Life Cycle.
7. What is Ab-initio Resource Assignment?
8. Define Periodic Resource Optimization.
9. What is meant by federated cloud computing?

10. State two functional interfaces of cloud services.

Understanding Level Questions (6 Marks each)

1. Explain the role of service providers and users in cloud computing.
2. Describe the architecture of federated cloud computing with a neat diagram.
3. Differentiate between the types of Service Level Agreements.
4. Explain the life cycle stages of an SLA in detail.
5. Describe the purpose and contents of a service catalog.
6. Discuss the management and functional interfaces used in cloud services.
7. Explain any four functions of a cloud portal.
8. Describe each phase of the cloud service life cycle with examples.
9. Explain Ab-initio Resource Assignment with suitable examples.
10. Describe the concept of Periodic Resource Optimization and its importance.

Application Level Questions (4 Marks each)

1. Suppose you are a cloud service provider, how will you ensure an effective SLA with your clients?
2. Design a simple federated cloud architecture for a university collaboration system.
3. Prepare a basic service catalog for an online storage cloud service.
4. Draft an example of a service planning phase for a cloud-based food delivery application.
5. Identify functional interfaces required for integrating a payment gateway in a cloud platform.
6. Assume you are managing a cloud portal, list and describe the functions you would prioritize.
7. Apply the service operation phase to a real-world cloud service like Google Drive.
8. Create a small example to explain Ab-initio Resource Assignment in a multi-cloud environment.
9. Prepare a strategy for periodic resource optimization for a video streaming cloud service.
10. Formulate a simple SLA for a cloud backup service, mentioning key points.

Unit5

Remembering Level Questions (2 Marks each)

1. Define cloud security.
2. What are cloud risks?
3. List two types of cloud risks.
4. What are polity and organizational risks?
5. Define technical risks in cloud computing.
6. What are legal risks in cloud environments?
7. Name two technologies used for data security.
8. What is digital identity management?
9. Define content level security.

10. What is meant by Security-as-a-Cloud-Service?

Understanding Level Questions (6 Marks each)

1. Explain the fundamentals of cloud security.
2. Describe the division of cloud risks with examples.
3. Differentiate between polity/organizational risks and technical risks.
4. Explain how legal risks impact cloud service providers.
5. Describe major technologies used for securing data in cloud computing.
6. What are data security risks? Explain with suitable examples.
7. Explain the role of digital identity and access management in cloud security.
8. Describe content-level security and its importance in the cloud.
9. Discuss the concept and benefits of Security-as-a-Cloud-Service.
10. Explain the steps involved in managing digital identities in a cloud environment.

Application Level Questions (4 Marks each)

1. Identify and explain possible cloud security fundamentals for an online banking cloud system.
2. Categorize cloud risks for a healthcare cloud platform (Polity, Technical, Legal).
3. Suggest technologies for securing customer data in an e-commerce cloud application.
4. Propose a solution for managing data security risks in a public cloud environment.
5. Create a strategy for digital identity and access management for a university's cloud services.
6. Apply content-level security measures for a video-on-demand cloud platform.
7. Design a basic plan for offering Security-as-a-Cloud-Service to small businesses.
8. Draft an example where a legal risk impacts cloud usage by an international organization.
9. Recommend technologies that can minimize technical risks in hybrid cloud systems.
10. Illustrate how a company can implement Security-as-a-Cloud-Service for its HR portal.